

Major Advanced Topics

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Strong-Field Ionization Phenomena Revealed by Quantum Trajectories

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Laser-atom interaction in Strong field

■ The Keldysh parameter

$$\gamma_K \equiv \omega \sqrt{2I_p} / E_0$$

$\gamma_k < 1 \longrightarrow$ Tunnel ionization (horizontal channel)

$\gamma_k > 1 \longrightarrow$ Tunnel ionization (vertical channel)

$\gamma_k \sim 1 \longrightarrow$ Intermediate regime

- Provides a quantitative understanding of the dominant photoionization mechanism
- Disregards the nature of the binding potential and excited state
- Reduces the intricacies of the field to the field strength only
- Wrongly assume that the tunneling wavepacket always start in the classically allowed region
- Over the barrier ionization (OBI) is poorly rendered

■ Since the dynamics of the electron becomes classical, trajectories based approaches are appropriate

■ This paper attempts at proposing a more suitable parameters than γ_K for describing the interaction

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The de Broglie-Bohm quantum theory

Trajectories are researched by solving the TDSE

$$i \frac{d}{dt} \Psi(t) = H(t) \Psi(t), \quad H(t) = H_0 - \mathbf{p} \mathbf{A}(t),$$
$$\mathbf{E}(t) = -\frac{d}{dt} \mathbf{A}(t), \quad \mathbf{A}(t) = -A_0 \cos^4 \left(\frac{\pi t}{NT} \right) \sin(\omega t) \hat{\mathbf{x}}, \quad t \in [-NT/2, NT/2], \quad T = 2\pi/\omega$$

■ Bohmian formalism and quantum potential

$$i \frac{\partial \psi}{\partial t} = \left(-\frac{1}{2} \nabla^2 + V \right) \psi, \quad \text{with} \quad \psi = R \exp(iS), \quad \rho(\mathbf{r}, t) = |\psi(\mathbf{r}, t)|^2$$

Using the above expression of the wavefunction ψ in the Schrödinger equation, and considering only the real part leaves us with

$$\frac{\partial S}{\partial t} = -\frac{1}{2} (\nabla S)^2 + \frac{1}{2} \left(\frac{\nabla^2 R}{R} \right) - V$$

$$Q = -\frac{1}{2} \rho^{-1/2} \nabla^2 \rho^{1/2}$$

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Bohmian theory - Velocity field and trajectories

- In the Bohmian formalism, velocity is given by

$$\mathbf{v}(\mathbf{r}, t) = \mathbf{j}(\mathbf{r}, t) / \rho(\mathbf{r}, t), \quad \mathbf{j}(\mathbf{r}, t) = \text{Re} \left(\Psi^*(\mathbf{r}, t) (-i\Delta + \mathbf{A}(t)) \Psi(\mathbf{r}, t) \right)$$

- Retrieving ψ from the TDSE
 - Propagation of the trajectories from different positions $\mathbf{r}_0$
 - Reconstruction of the wavefunction from the trajectories at any given time
- Computation of time and radii exit t_{ex} and r_{ex} from the condition
$$\Delta \mathcal{E}(t_{\text{ex}}) = 0, \quad \Delta \mathcal{E}(t) = v^2(t)/2 + Q(t),$$
where $\Delta \mathcal{E}(t)$ is the difference between the energy of the trajectory $\mathcal{E}(t)$ and I_p
 - The quantum potential $Q(t)$ is responsible for all quantum effects and the nonzero velocity at the exit
 - In the classical limit

$$Q(t) \longrightarrow 0$$

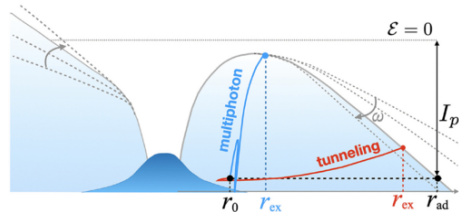
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The tunneling parameter

$$\beta(\mathbf{r}_0) \equiv \frac{r_{\text{ex}}(\mathbf{r}_0) - r_0}{r_{\text{ad}}(\mathbf{r}_0) - r_0}$$

- $0 \leq \beta \leq 1$ quantifies the degree of tunneling for a trajectory starting at r_0
 - For vertical ionization, $r_{\text{ex}} = r_0 \rightarrow \beta = 0$
 - For horizontal ionization, $r_{\text{ex}} = r_{\text{ad}} \rightarrow \beta = 1$
 - β also exists in the eventuality that $r_{\text{ex}} \geq r_{\text{ad}}$ (OBI...)
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- The adiabatic radius r_{ad} corresponds to a trajectory such as $\mathcal{E} = -I_p$



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Photoionization of atomic hydrogen as a function of r_0

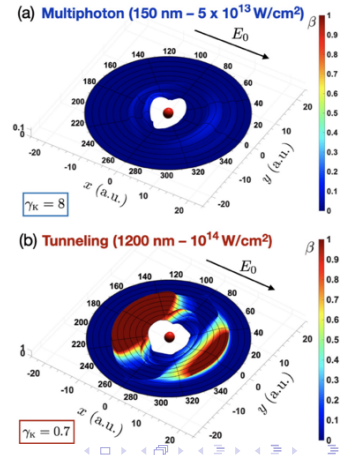
- 2 dimensional plot of $\beta(r_0)$ as a function of r_0 in the xy -plane
- 2 different pulses describing MPI and TI are used
- Fig. 2(b) provides additional details on the dynamics of the ionization
 - Tunnel ionization dominates in the direction longitudinal to the field past a certain r_0
 - Multiphoton mechanism occurs in the direction transversal to the field ionization
 - At large distances, the mechanism naturally exhibits a multiphoton character

$$\beta \longrightarrow 0$$

vertical channel

$$\beta \longrightarrow 1$$

horizontal channel



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Quantum adiabatic parameter

- Statistically averaged tunneling level of the ionization process

$$\bar{\beta} = W_0^{-1} \int_{\mathcal{D}} \beta(\mathbf{r}_0) \rho(\mathbf{r}_0, 0) d^3\mathbf{r}_0$$

- $\mathcal{D}$ denotes the set of ionizing trajectories
- $W_0 = \int_{\mathcal{D}} \rho(\mathbf{r}_0, 0) d^3\mathbf{r}_0$ is the total ionization probability

- To compare with the Keldysh parameter γ_K , they define the quantum adiabatic parameter γ_q

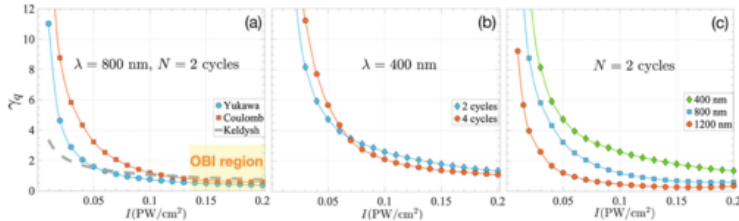
$$\gamma_q \equiv \frac{1 - \bar{\beta}}{\bar{\beta}}$$

- This quantity has the same limits and meaning as γ_K
- $\gamma_q \rightarrow 0$ for TI and $\gamma_q \rightarrow \infty$ for MPI

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Quantum adiabatic parameter as a function of the intensity



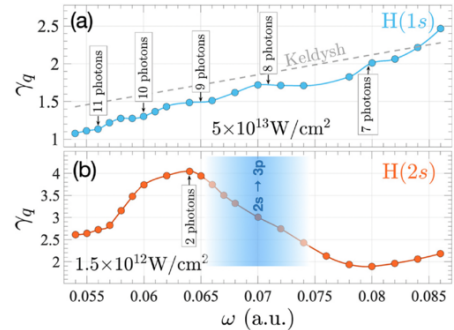
- Fig.3(a) Comparison of γ_q for 2 different potentials with γ_K for different intensities
 - For weak fields, MPI is more pronounced than predicted by the Keldysh approximation
 - The difference between the potentials is sharp at low intensities. γ_q is more sensitive than γ_K
- Fig. 3(b) Comparison of γ_q for different pulse duration
 - 4 cycles has more enhanced MPI and TI mechanisms because more laser peaks
- Fig. 3(c) Comparison of γ_q for different wavelengths
 - The results agree with general prediction

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Quantum adiabatic parameter as a function of the frequency

- Fig. 4(a) Intermittent steps at which γ_q ceases to progress uniformly are observed
 - Correspond to channel closings known to suppress ionization
- Fig. 4(b) Results for 2 photon ionization in the $2s$ state
 - Sharp drop near resonance indicating a preference for TI in this frequency region
 - Population transfer to the $3p$ state favoring TI
 - The point is that it reproduces a specific behaviour reported in the literature¹



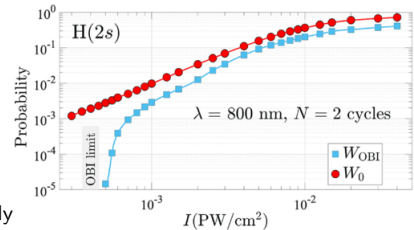
¹K. Mishima et al., Phys. Rev. A 66, 033401 (2022)

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Over the barrier ionization

- OBI happens if either $\mathcal{E}(t)$ remains above the barrier or $\mathcal{E}(t_{\text{ex}}) \leq I_p$
- Going by this definition, Fig. 5 compares W_0 and W_{OBI} for different intensities
 - W_{OBI} experiences a sharp increase when the top of the barrier becomes lower than I_p
 - Subsequently, the increase rate is similar to W_0
- At intensities about 10 times higher than the OBI limit, a steady state is reached where $W_{\text{OBI}}/W_0 \sim 0.6$



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Closure

- Quantum trajectories represent a powerful alternative tool to investigate electron dynamics beneath and above the potential barrier
- Rare quantitative information can be probed about various high intensity effects
- Importance of long range electronic potential and nonadiabatic effects have been shown
- This might pave the way to a better understanding of other routinely studied dynamical effects (HHG, etc..)